

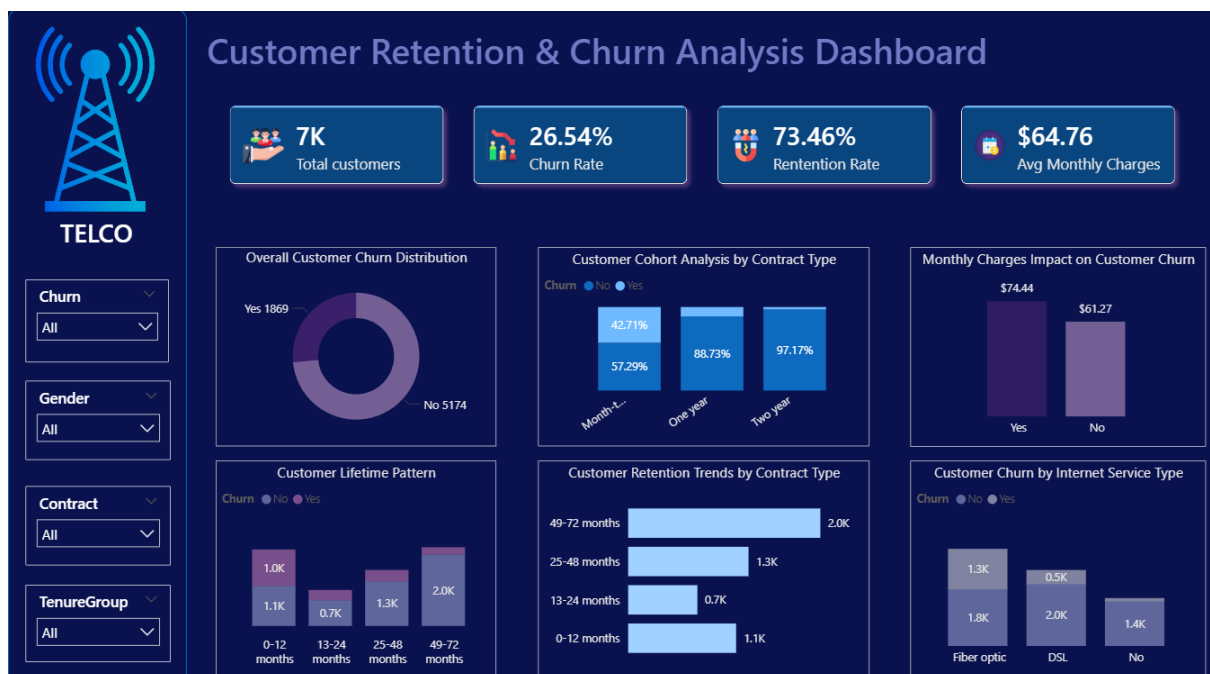
CUSTOMER RETENTION & CHURN ANALYSIS

Data Science & Analytics Internship Project

Prepared By: Hema Malini

Tools Used: Python & Power BI

Dataset: Telco Customer Churn Dataset (Kaggle)



Customer Retention & Churn Analysis Report

1. Executive Summary

The telecom company serves **7,043 customers** with a retention rate of **73.46%** and a churn rate of **26.54%**. The analysis shows that churn is highest among month-to-month customers, customers with shorter tenure, and those paying higher monthly charges. Long-term contracts and longer customer relationships contribute significantly to retention.

2. Business Objective

- Identify key reasons for customer churn.
 - Analyse retention trends and customer behaviour.
 - Find high-risk customer segments.
 - Recommend strategies to improve retention and reduce churn.
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3. Dataset Overview

The analysis was performed using the **Telco Customer Churn Dataset** sourced from [Kaggle](#).

The dataset contains customer subscription and service-related information, including:

- Customer demographics
- Contract type
- Tenure group
- Monthly charges
- Internet service type
- Churn status

Key Metrics

- Total Customers: **7,043**
- Churn Rate: **26.54%**
- Retention Rate: **73.46%**
- Avg Monthly Charges: **\$64.76**

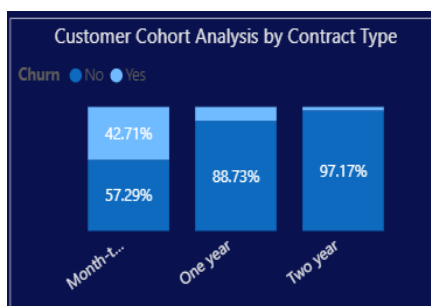
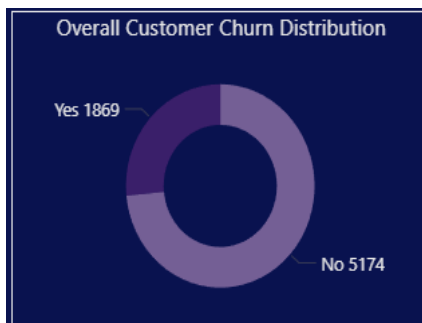
4. Tools Used

- Python (Data Cleaning & Analysis)
 - Microsoft Power BI (Dashboard & Visualisation)
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5. Dashboard & Visuals

The dashboard includes:

- KPI Summary Cards
- Churn Distribution
- Customer Lifetime Patterns & cohort analysis



6. Analysis & Findings

- Month-to-month customers have the highest churn.
 - Two-year contract customers show the strongest retention.
 - Churned customers pay higher monthly charges (\$74.44).
 - Customers within 0–12 months are more likely to leave.
 - Retention improves as customer tenure increases.
 - Fiber optic users show higher churn compared to DSL users.
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7. Recommendations

- Improve onboarding for new customers.
 - Encourage long-term contracts through offers and discounts.
 - Review pricing for high-charge customers.
 - Improve fiber optic service quality and support.
 - Implement targeted retention campaigns for high-risk customers.
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8. Business Impact

- Better onboarding may reduce early churn.
 - Long-term contracts can improve revenue stability.
 - Pricing optimisation may increase customer satisfaction.
 - Improved service quality can strengthen customer loyalty.
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9. Conclusion

The business maintains strong retention overall, but churn remains high among short-term and high-charge customers. Focusing on customer experience, pricing, and long-term engagement strategies will help improve retention and support sustainable business growth.